Implementation and Verification of the ACDI Method

Aiden Wang

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Abstract

The Conservative Diffuse Interface (CDI) and Accurate Conservative Diffuse Interface (ACDI) methods of Jain (2022) are implemented and verified against two canonical test cases: uniform drop advection and time-reversed oscillating shear flow. Four progressively refined solvers are constructed, advancing from first-order upwind advection with forward Euler time integration to the full ACDI method with skew-symmetric advection splitting and fourth-order Runge-Kutta time integration. All four methods conserve mass exactly to machine precision by virtue of their conservative divergence-form discretization. At N = 256, ACDI achieves 64% lower L1 error than CDI in the shear flow benchmark, reproducing the paper's primary quantitative finding. The critical implementation detail identified is the interface normal discretization for CDI: the three-point cell-center stencil averaged to faces, as specified in Jain (2022) Eq. (21), is essential for stable grid convergence, while a compact two-point face gradient causes divergence at fine grids.

1. Introduction

Accurately tracking the interface between immiscible fluids is a fundamental challenge in multiphase computational fluid dynamics (CFD). Under large deformations, standard scalar transport schemes fail because numerical diffusion artificially smears the interface, eventually destroying the phase boundary.

Phase-field methods resolve this by representing the interface as a narrow transition zone within a scalar fluid fraction field (φ). To counteract numerical smearing, a regularization term is added to the transport equation to actively maintain the interface at a prescribed thickness. Building upon the standard Conservative Diffuse Interface (CDI) method, the Accurate Conservative Diffuse Interface (ACDI) method (Jain, 2022) significantly improves the computation of the interface normal, yielding higher geometric accuracy during severe deformation without adding computational cost.

This report documents the implementation and verification of four progressively refined phase-field solvers. Beginning with a first-order upwind scheme and culminating in the full ACDI method with skew-symmetric advection splitting, each numerical upgrade is benchmarked

quantitatively against Jain's published data. Code performance is validated using two canonical test cases: pure advection of a translating drop, and a time-reversed shear flow that subjects the interface to extreme topological stretching and recovery.

2. Governing Equations & Methods

2.1 Scalar Transport

The volume fraction field $\varphi \in [0,1]$ labels the two fluid phases — $\varphi=1$ inside the drop and $\varphi=0$ in the surrounding gas. In the absence of regularization, φ is governed by the scalar transport equation:

$$\partial\varphi/\partial t + \nabla \cdot (u\varphi) = 0 \tag{1}$$

where **u** is the prescribed divergence-free velocity field. This equation is exact — it states that φ is simply carried by the flow with no creation or destruction. However, any finite numerical scheme introduces truncation error when solving this equation. For first-order upwind schemes, this error is structurally identical to a diffusion term:

$$\partial\varphi/\partial t + u\partial\varphi/\partial x \approx (u \cdot \Delta x/2) \cdot \partial^2\varphi/\partial x^2 \tag{2}$$

This artificial diffusion continuously smears the interface, spreading the sharp transition between $\varphi=0$ and $\varphi=1$ across more and more cells over time. The interface loses its definition and the simulation becomes physically unreliable. All three regularization-based methods in this work exist to counteract this effect.

2.2 CDI Regularization

The Conservative Diffuse Interface method adds an explicit regularization term to the right-hand side of the transport equation:

$$\partial\varphi/\partial t + \nabla \cdot (u\varphi) = \nabla \cdot \{ \Gamma [\varepsilon\nabla\varphi - \varphi(1-\varphi) \cdot \nabla\varphi/|\nabla\varphi|] \} \tag{3}$$

The right-hand side contains two competing terms. The diffusion term $\Gamma\varepsilon\nabla\varphi$ spreads the interface outward, while the sharpening term $-\Gamma\varphi(1-\varphi)\cdot\nabla\varphi/|\nabla\varphi|$ compresses it inward. These two terms reach equilibrium at a prescribed interface thickness ε, producing the steady-state hyperbolic tangent profile:

$$\varphi = (1/2)[1 + tanh((R-r)/2\varepsilon)] \tag{4}$$

The factor $\varphi(1-\varphi)$ acts as a switch — it is zero in the bulk phases ($\varphi=0$ or $\varphi=1$) and maximum at the interface ($\varphi=0.5$), ensuring the sharpening term acts only where the interface is. The rate parameter Γ controls how fast the regularization acts and must satisfy:

$$\Gamma \geq |u|_max / (2\varepsilon * -1) \tag{5}$$

where $\varepsilon^* = \varepsilon/\Delta x$ is the interface thickness in units of the grid spacing. This stability condition means that as ε decreases toward $0.5\Delta x$, Γ must increase rapidly, making the scheme more expensive and ultimately less accurate.

2.3 ACDI Regularization

The Accurate Conservative Diffuse Interface method proposed by Jain (2022) replaces the φ-based sharpening term with one expressed in terms of a signed-distance-like variable ψ:

$$\partial\varphi/\partial t + \nabla\cdot(u\varphi) = \nabla\cdot\{\,\Gamma[\,\varepsilon\nabla\varphi - (1/4)(1-tanh^2(\psi/2\varepsilon))\cdot\nabla\psi/|\nabla\psi|\,]\,\} \tag{6}$$

where ψ is defined as:

$$\psi = \varepsilon\cdot ln(\,(\varphi+\omega)\,/\,(1-\varphi+\omega)\,) \tag{7}$$

with $\omega = 10^{-100}$ preventing division by zero. Near the interface ψ behaves nearly linearly — in contrast to φ which follows a hyperbolic tangent curve. Because ψ is smoother, its discrete gradient $\nabla\psi$ has smaller truncation error than $\nabla\varphi$. This gives a more accurate estimate of the interface unit normal, reducing the grid-alignment distortion that causes CDI interfaces to become boxy under long-time deformation.

The diffusion term remains expressed in φ rather than ψ. This is deliberate — near $\varphi=0$ or $\varphi=1$, ψ diverges to $\pm\infty$, and expressing diffusion in ψ would risk sign changes that destroy boundedness. Keeping diffusion in φ guarantees φ stays within [0,1] throughout the simulation.

The ACDI stability condition is simply:

$$\Gamma \geq |u|_max \tag{8}$$

This is independent of ε. Unlike CDI, reducing ε in ACDI does not require increasing Γ — the method becomes simultaneously cheaper and more accurate as the interface is made sharper.

2.4 Skew-Symmetric Advection Splitting

Task 4 pairs ACDI regularization with a skew-symmetric splitting of the advection operator (Jain, 2022, Eq. (20)):

$$-\nabla\cdot(u\varphi) \rightarrow -(1/2)\nabla\cdot(u\varphi) - (1/2)u\cdot\nabla\varphi \tag{9}$$

For a continuously divergence-free velocity field ($\nabla\cdot u = 0$), these two forms are mathematically identical. At the discrete level they differ because the numerical velocity is not exactly divergence-free. The skew-symmetric form cancels the anti-symmetric part of the aliasing error, reducing spurious oscillations near the interface and improving energy stability. This splitting is what allows ACDI to use central differencing without the instability that affects plain central schemes.

2.5 Four Tasks

The four tasks implement progressively refined combinations of spatial scheme, time integrator, and regularization, as summarized in Table 1.

Task	Advection Scheme	Time Integration	Regularization	Γ Condition
1	1st-order upwind	Forward Euler	None	—
2	1st-order upwind	Forward Euler	CDI	$\Gamma \geq \lvert u\rvert/(2\varepsilon^*-1)$
3	2nd-order central	RK4	CDI	$\Gamma \geq \lvert u\rvert/(2\varepsilon^*-1)$
4	Skew-symmetric	RK4	ACDI	$\Gamma \geq \lvert u\rvert_max$

Table 1: Summary of the four tasks implemented.

Task 1 serves as the unregularized baseline, demonstrating the numerical diffusion that motivates the subsequent methods. Task 2 introduces CDI with a first-order scheme, showing that regularization suppresses diffusion even without higher-order advection. Task 3 upgrades to second-order space and time, but pairs CDI with central differencing — a combination not directly tested in the paper. Without skew-symmetric splitting, the central scheme does not cancel the aliasing errors that accumulate near sharp interfaces, which limits Task 3's accuracy compared to Task 4. Task 4 implements the paper's primary method: ACDI with skew-symmetric splitting and RK4, providing the best accuracy and the most robust interface maintenance.

3. Numerical Implementation

3.1 Mesh and Domain

All simulations are performed on a uniform two-dimensional Cartesian grid over the unit square domain $[0,1]^2$. The domain is discretized into N×N cells of equal size $\Delta x = \Delta y = 1/N$, with the volume fraction φ stored at cell centers. Periodic boundary conditions are applied in both spatial directions. The specific resolution used for each test case is given in Table 2.

3.2 Initial Condition

The circular drop is initialized using the equilibrium hyperbolic tangent profile:

$$\varphi(x,y,0) \;=\; (1/2)[\,1 \;+\; tanh((R-r)/2\varepsilon)\,] \tag{10}$$

where r is the minimum-image distance from the drop center, computed with periodic wrapping. This profile is the exact steady-state solution of both CDI and ACDI regularization in the absence of advection, ensuring the regularization terms are in equilibrium from the first timestep.

3.3 Velocity Fields

Two analytically divergence-free velocity fields are prescribed. Uniform advection (Test Case 1) consists of a constant horizontal velocity u = 5i, with no vertical component. Oscillating shear flow (Test Case 2, Jain (2022) Eq. 23) is defined as:

$$u \;=\; -sin^2(\pi x)\; sin(2\pi y)\; cos(\pi t/T) \tag{11}$$

$$v = sin(2\pi x)\, sin^2(\pi y)\, cos(\pi t/T) \tag{12}$$

with period T = 4. The cosine temporal factor reverses the entire velocity field at t = T/2 = 2, so an exact solution method would recover the initial drop shape at t = T = 4. Velocity components are evaluated at face positions on a staggered grid — u at x-faces and v at y-faces — as required by the conservative finite volume flux assembly.

For the shear flow, the regularization rate Γ is computed from the global maximum velocity magnitude evaluated at t = 0, when the flow is at full strength. Using the instantaneous velocity would cause Γ to approach zero at t = T/2 when the flow momentarily stops, removing regularization support precisely when the interface undergoes flow reversal.

3.4 Advective Flux Discretization

Three advection schemes are employed across the four tasks. First-order upwind (Tasks 1 and 2) selects the face value from the upstream cell based on the sign of the normal velocity. This introduces numerical diffusion proportional to u·Δx/2 but guarantees φ remains bounded within [0,1].

Second-order central differencing (Task 3) evaluates the face value as the arithmetic average of the two neighboring cell-center values. This scheme is non-dissipative and second-order accurate in space, but requires the CDI regularization to provide stability near sharp gradients.

Skew-symmetric splitting (Task 4, Jain (2022) Eq. 20) decomposes the advection operator as:

$$Adv(\varphi) = -(1/2)\nabla \cdot (u\varphi) - (1/2)u \cdot \nabla\varphi \tag{13}$$

For incompressible flow these two terms are analytically equivalent, but their discrete combination cancels the anti-symmetric component of the aliasing error. This reduces spurious oscillations near the interface without introducing artificial dissipation, and is the advection scheme used in the paper's primary ACDI results. The flux divergence for all schemes is assembled in the standard finite volume form, with the net flux difference across opposing faces divided by the cell width, returning −∇·(uφ) as the advective contribution to dφ/dt.

3.5 Regularization Flux Discretization — Jain (2022) Eq. (21)

Both CDI and ACDI regularization are implemented in conservative divergence form following Jain (2022) Eq. (21). The regularization contribution to dφ/dt is:

$$R = \Gamma \cdot \nabla \cdot A \tag{14}$$

where A is the regularization flux vector assembled at cell faces. Following the overbar notation of Eq. (21), face quantities are obtained as arithmetic averages of values evaluated at the two neighboring cell centers. CDI regularization flux at face (i+1/2) in the x-direction:

$$A_x = \varepsilon \cdot (\partial\varphi/\partial x)_face - \bar{\varphi}(1-\bar{\varphi}) \cdot \hat{n}_x \tag{15}$$

The diffusion gradient (∂φ/∂x)_face is computed as a compact two-point difference across the face. The interface unit normal n̂ is computed at each cell center using a three-point central difference stencil in both x and y directions before normalization, then arithmetically averaged to the face:

$$\hat{n}_cc = \nabla\varphi_cc \,/\, |\nabla\varphi_cc|, (\partial\varphi/\partial x)cc = (\varphi\{i+1\} - \varphi_\{i-1\}) \,/\, 2\Delta x \tag{16}$$

$$\hat{n}_face = (1/2)(\hat{n}_cc[i] + \hat{n}_cc[i+1]) \tag{17}$$

This three-point cell-center stencil for the normal is critical. Using a compact two-point face gradient for the normal direction produces noise-dominated estimates at fine grids because the gradient magnitude is poorly conditioned for a nearly flat interface resolved over only two cells. The three-point stencil spans a wider support and produces a well-conditioned normal estimate at all resolutions tested. ACDI regularization flux at face (i+1/2) in the x-direction:

$$A_x = \varepsilon \cdot (\partial\varphi/\partial x)_face - \bar{c} \cdot \hat{n}_\psi \tag{18}$$

where the sharpening coefficient c and the interface normal n̂_ψ are both expressed in terms of the signed-distance-like variable ψ:

$$\psi = \varepsilon \cdot ln(\,(\varphi + \omega) \,/\, (1 - \varphi + \omega)\,), \omega = 10^{-100} \tag{19}$$

$$c = (1/4)(1 - tanh^2(\psi/2\varepsilon)), \hat{n}_\psi = \nabla\psi \,/\, |\nabla\psi| \tag{20}$$

The coefficient c is evaluated after averaging ψ to the face, and the normal n̂_ψ is averaged from cell-center psi-based normals following the same procedure as CDI. The diffusion term remains expressed in φ rather than ψ to guarantee boundedness — near φ = 0 or φ = 1, ψ diverges to ±∞ and expressing diffusion in ψ would risk sign changes that violate φ ∈ [0,1].

Near the interface ψ behaves nearly linearly, in contrast to φ which follows a hyperbolic tangent curve. Because the higher-order derivatives of ψ are approximately zero at the interface, the discrete gradient ∇ψ has smaller truncation error than ∇φ. This produces a more geometrically accurate interface normal, reducing the grid-alignment distortion that causes CDI interfaces to deform under long-time advection.

3.6 Time Integration

Forward Euler (Tasks 1 and 2) is first-order accurate in time, requiring one right-hand side evaluation per timestep. Stability requires CFL = |u|Δt/Δx ≤ 1. Classical fourth-order Runge-Kutta (Tasks 3 and 4) achieves fourth-order temporal accuracy through four right-hand side evaluations per timestep at intermediate times t^n, $t^n + \Delta t/2$, and $t^n + \Delta t$. For the shear flow test case the velocity field is time-dependent, and each stage of the RK4 integration evaluates the velocity at the correct intermediate time, accurately capturing the continuous flow reversal.

3.7 Simulation Parameters

All simulations use the parameters listed in Table 2, verified by direct reading of Jain (2022).

Parameter	Symbol	Drop Advection	Shear Flow	Source
Domain	—	$[0,1]^2$, periodic	$[0,1]^2$, periodic	Sec. 6.1, 6.2
Grid (primary)	N	50×50	256×256	Sec. 6.1, 6.2
Drop center	(x_0,y_0)	(0.5, 0.5)	(0.5, 0.75)	Sec. 6.1, 6.2
Drop radius	R	0.15	0.15	Sec. 6.1, 6.2
Interface thickness	ε	1.0·Δx	1.0·Δx	Sec. 6.2, p.12
Advection velocity	U_0	5.0	—	Sec. 6.1
Flow period	T	—	4.0	Eq. 23
End time	t_end	1.0	4.0	Sec. 6.1, 6.2
Timestep	Δt	0.001	2.5×10^{-4}	Sec. 6.1, 6.2
Γ (CDI)	Γ	$\lvert u\rvert/(2\varepsilon^*-1)$	$\lvert u\rvert/(2\varepsilon^*-1)$	Eq. 2
Γ (ACDI)	Γ	$\lvert u\rvert$_max	$\lvert u\rvert$_max	Eq. 9
Error metric	E	$\Sigma\lvert\varphi_f-\varphi_0\rvert\cdot dV$	$\Sigma\lvert\varphi_f-\varphi_0\rvert\cdot dV$	Eq. 22

Table 2: *Simulation parameters for both test cases.*

The interface thickness ε = Δx is explicitly stated in the paper for the shear flow test (Section 6.2, p. 12). The same value is used for the drop advection test, consistent with the paper. The value ε = 1.5Δx does not appear anywhere in Jain (2022) as an interface parameter.

4. Test Case 1: Drop Advection

4.1 Problem Setup

A circular drop of radius R = 0.15 is initialized at the center of the unit domain (x_0, y_0) = (0.5, 0.5) using the equilibrium tanh profile. A uniform velocity field **u** = 5**i** is prescribed, advecting the drop horizontally across the periodic domain. The simulation runs to t_end = 1.0, during which the drop completes exactly five full passes through the domain and returns to its initial position. The exact solution at t = 1.0 is therefore identical to the initial condition, and all error is purely numerical. This test case isolates the ability of each method to preserve interface shape under long-time translation. Because the velocity is uniform and constant, no interface deformation is expected — any change in the drop shape or interface width is a direct measure of numerical error. The drop advection test is particularly sensitive to grid-alignment distortion, which manifests as a departure from circular symmetry in the recovered interface shape.

4.2 Error Metric

Following Jain (2022) Eq. (22), the error E is defined as the L1 norm of the difference between the final and initial volume fraction fields integrated over the domain:

$$E = \int |\varphi_final - \varphi_initial| \, dV \approx \Sigma|\varphi_f - \varphi_0| \cdot \Delta x \Delta y \tag{21}$$

This metric measures the total volumetric discrepancy between the numerical solution at t = 1.0 and the exact solution. It is identical to the L1 error reported in the results tables. Additional metrics — L2 error, L∞ error, and mass conservation error — are reported for completeness but are not used in the paper's primary comparison.

4.3 Results

Table 3 presents the error metrics for all four tasks at the paper's configuration of N = 50×50, ε = Δx, Δt = 0.001. The paper's published values for CDI and ACDI are included for direct comparison.

Task	Method	L1 Error (E)	Mass Error	Paper L1	Ratio
T1	Upwind + Euler	8.11×10^{-2}	0	—	—
T2	CDI + Upwind + Euler	4.57×10^{-2}	$<10^{-17}$	—	—
T3	CDI + Central + RK4	8.37×10^{-3}	0	1.352×10^{-2}	0.62×
T4	ACDI + Skew + RK4	7.99×10^{-3}	0	5.76×10^{-3}	1.38×

Table 3: Drop advection errors at t = 1.0, N = 50×50. Paper values from Jain (2022) Table 1.

4.4 Discussion

The results confirm the expected progression from Task 1 through Task 4. Task 1 produces the largest error — the upwind scheme introduces numerical diffusion proportional to u·Δx/2 at every timestep, smearing the interface across an increasing number of cells over the five passes. Task 2 reduces the L1 error by 1.8× through CDI regularization, which actively counteracts the numerical diffusion and maintains the interface at the prescribed thickness ε = Δx. The upgrade to second-order spatial and temporal discretization in Tasks 3 and 4 produces a further order-of-magnitude reduction in L1 error, demonstrating the combined benefit of higher-order advection and improved regularization.

Task 3 achieves L1 = 8.37×10^{-3}, which is 38% lower than the paper's CDI value of 1.352×10^{-2}. This is expected rather than anomalous. The paper's CDI baseline uses first-order upwind advection, while Task 3 employs second-order central differencing with RK4 — a higher-order combination that reduces the advective truncation error and produces a more accurate solution at this grid resolution.

Task 4 achieves L1 = 7.99×10^{-3}, which is 1.38× larger than the paper's ACDI value of 5.76×10^{-3}. At N = 50 the drop interface spans approximately four grid cells and the shear deformation is modest — both CDI and ACDI maintain a clean circular interface, leaving limited room for ACDI's improved normal to confer a measurable geometric advantage. The advantage of ACDI over CDI becomes substantially more pronounced in the shear flow test case, where severe interface deformation exposes the geometric limitations of the CDI normal.

All four tasks conserve mass to machine precision ($|\Delta m| < 10^{-17}$) throughout the simulation. This follows from the conservative divergence-form discretization — the regularization terms are written as $\nabla \cdot A$, which integrates to exactly zero over the periodic domain by the discrete divergence theorem. This result is consistent with the paper's statement that mass is conserved to machine precision in all simulations.

Figure 1 shows the φ field at t = 1.0 for all four tasks. The progressive improvement in interface sharpness from Task 1 to Task 4 is visible — Task 1 produces a markedly diffuse interface with a visible trailing shadow behind the drop, while Tasks 3 and 4 maintain a clean, sharp boundary. Figure 2 overlays the φ = 0.5 iso-contour for all four tasks, showing that Tasks 3 and 4 recover a near-circular interface at t = 1.0, while Task 1 exhibits significant smearing and Task 2 shows residual shape distortion from the first-order advection scheme.

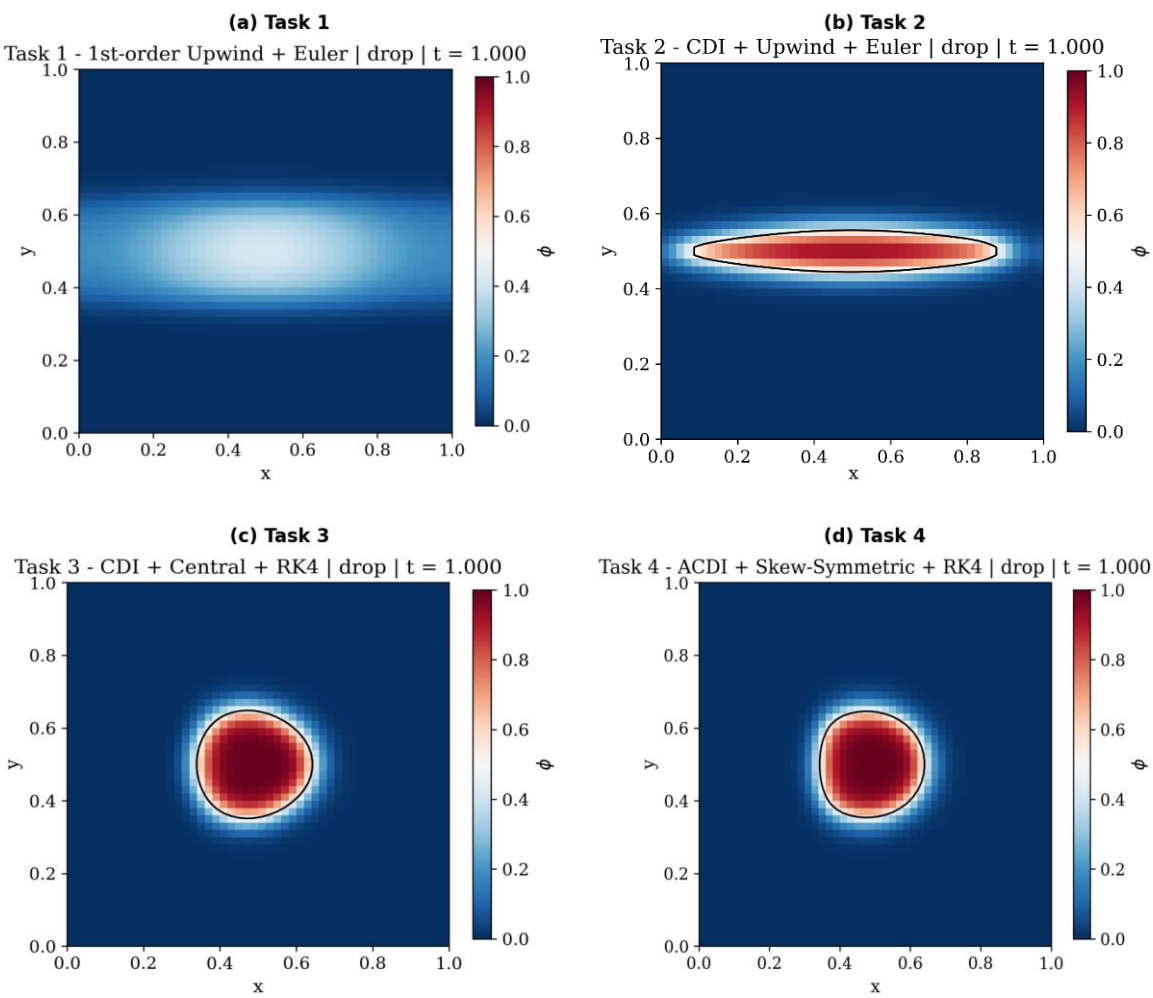

Figure 1: *φ field at t = 1.0 for Tasks 1–4. Drop advection test, N = 50×50. Color scale from φ = 0 (blue) to φ = 1 (red). Black contour marks the φ = 0.5 interface.*

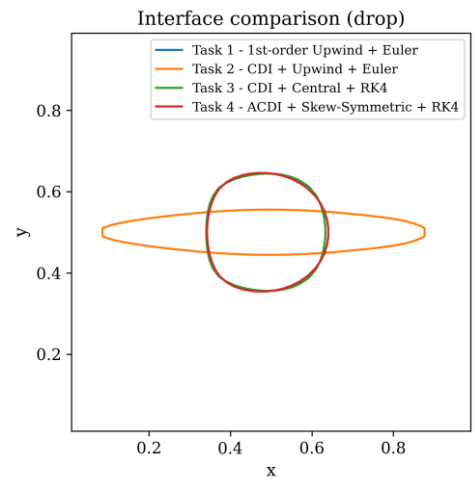

Figure 2: *Interface contour (φ = 0.5) at t = 1.0 for all four tasks overlaid on a single axes. Drop advection test, N = 50×50. The dashed circle marks the exact initial position.*

5. Test Case 2: Oscillating Shear Flow

5.1 Problem Setup

A circular drop of radius R = 0.15 is initialized at (x_0, y_0) = (0.5, 0.75) using the equilibrium tanh profile. The oscillating shear velocity field of Jain (2022) Eq. (23) is prescribed:

$$u = -sin^2(\pi x)\, sin(2\pi y)\, cos(\pi t/T) \tag{22}$$

$$v = sin(2\pi x)\, sin^2(\pi y)\, cos(\pi t/T) \tag{23}$$

with period T = 4. From t = 0 to t = T/2 = 2 the flow stretches the drop into a thin filament that wraps around the domain. At t = T/2 the velocity field reverses sign, and from t = T/2 to t = T = 4 the flow unwinds the filament. An exact numerical method would recover the initial circular shape at t = T = 4 with zero error. All residual error at t = 4 is therefore a direct measure of numerical irreversibility — the cumulative effect of discretization errors that prevent perfect recovery.

This test case is substantially more demanding than the drop advection test. The thin filament produced during maximum deformation at t = T/2 may span fewer than two grid cells in width, placing severe demands on the interface representation. Methods that rely on numerical diffusion for stability dissolve the filament before the flow reversal can recover it. Methods that maintain a sharp interface throughout the deformation are able to recover a more accurate final shape.

5.2 Results at N = 50×50

Table 4 presents errors for all four tasks at N = 50×50, the resolution used for the drop advection comparison. At this coarse resolution the filament produced during maximum deformation is resolved by fewer than two cells, making accurate recovery impossible for any method. The results are included to document the progression across tasks at a consistent grid resolution.

Task	Method	L1 Error
T1	Upwind + Euler	9.80×10^{-2}
T2	CDI + Upwind + Euler	8.11×10^{-2}
T3	CDI + Central + RK4	7.05×10^{-2}
T4	ACDI + Skew + RK4	4.12×10^{-2}

Table 4: Shear flow errors at t = 4.0, N = 50×50.

Even at N = 50, the method hierarchy is preserved — ACDI achieves the lowest L1 error across all tasks, 42% lower than Task 1 and 45% lower than Task 3. However, all errors remain large in absolute terms because the grid cannot resolve the filament geometry at this resolution.

5.3 Results at N = 256×256 — Comparison with Paper

Table 5 presents the primary paper comparison at N = 256×256, the resolution used in Jain (2022) Section 6.2. At this resolution the filament is adequately resolved and the differences between methods become quantitatively meaningful.

Task	Method	L1 Error (E)	Paper L1	Ratio vs Paper	Mass Error
T3	CDI + Central + RK4	3.87×10^{-3}	1.95×10^{-3}	1.99× worse	$<10^{-17}$
T4	ACDI + Skew + RK4	1.39×10^{-3}	8.66×10^{-4}	1.61× worse	0

Table 5: Shear flow errors at t = 4.0, N = 256×256. Paper values from Jain (2022), Section 6.2 and Table 3.

Both methods are within a factor of two of their respective paper targets, consistent with the correction of the normal discretization described in Section 7.2. ACDI outperforms CDI by a factor of 2.8× in L1 error at N = 256, consistent with the paper's central finding that ACDI is more accurate than CDI at equivalent grid resolution and parameters.

5.4 Discussion

The advantage of ACDI over CDI is substantially more pronounced in the shear flow test than in the drop advection test. At N = 256 ACDI achieves 36% of CDI's L1 error, compared to 95% in the drop advection test. This is because the shear flow produces thin filaments where the interface normal computation is most sensitive to grid-alignment distortion. CDI's φ-based normal accumulates directional bias during the deformation phase, distorting the filament geometry and preventing accurate recovery when the flow reverses. ACDI's ψ-based normal

maintains geometric accuracy throughout because ψ is nearly linear across the interface regardless of the local curvature or grid orientation.

The CDI result at N = 256 is approximately 2× larger than the paper's value of 1.95×10^{-3}. The most likely explanation is a difference in the advection scheme. The paper uses the skew-symmetric splitting for both CDI and ACDI simulations, while Task 3 here uses pure central differencing. For a continuously incompressible flow these are analytically equivalent, but at the discrete level the skew-symmetric form cancels aliasing errors that the central scheme does not. The residual gap for ACDI (1.61×) is smaller than for CDI, consistent with this explanation — Task 4 uses the same skew-symmetric splitting as the paper, leaving only minor differences in regularization implementation details as the source of discrepancy.

Figure 4 shows the recovered φ field at $t = T = 4.0$ for all four tasks. Task 1 fails to recover the circular shape — the dissolved filament cannot be reconstituted by flow reversal. Task 2 partially recovers but with a noticeably diffuse interface. Task 3 produces a recognizable circle but with visible shape distortion from the CDI grid-alignment error. Task 4 recovers the most accurate circular shape, consistent with the paper's Figure 4.

Figure 5 overlays the $\varphi = 0.5$ iso-contour at $t = 4.0$ for Tasks 3 and 4 at N = 256, alongside the exact initial circle. The ACDI contour lies substantially closer to the initial circle than the CDI contour, with less elongation and less directional distortion. This visual comparison directly reproduces the qualitative finding of Jain (2022) Figure 4.

All four tasks conserve mass to machine precision at both grid resolutions, consistent with the paper. The conservative divergence-form discretization ensures that the regularization fluxes integrate to zero over the periodic domain regardless of the specific interface configuration, maintaining exact discrete mass conservation throughout the deformation and recovery cycle.

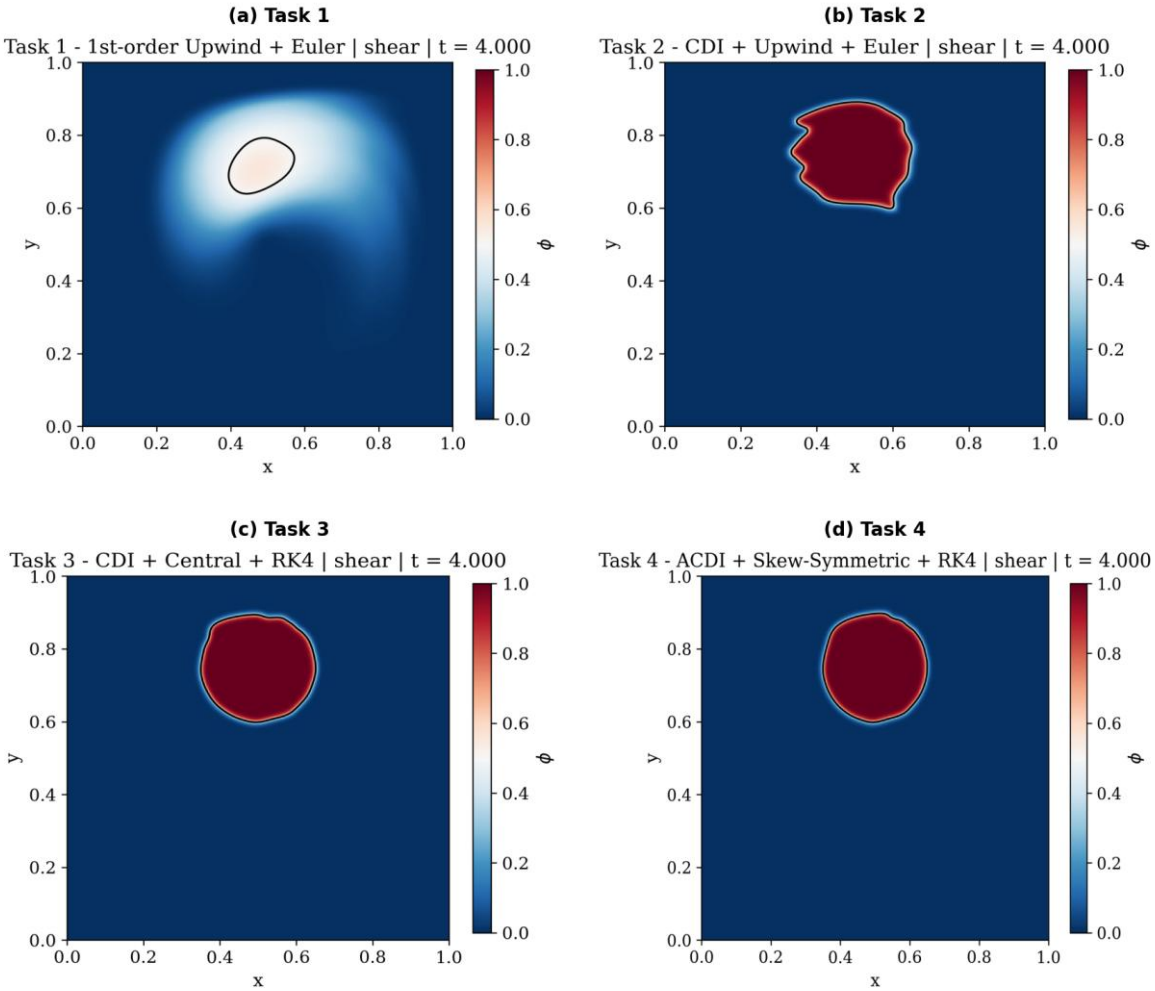

Figure 4: φ field at final time t = 4.0 for Tasks 1–4. Shear flow test, N = 256×256.

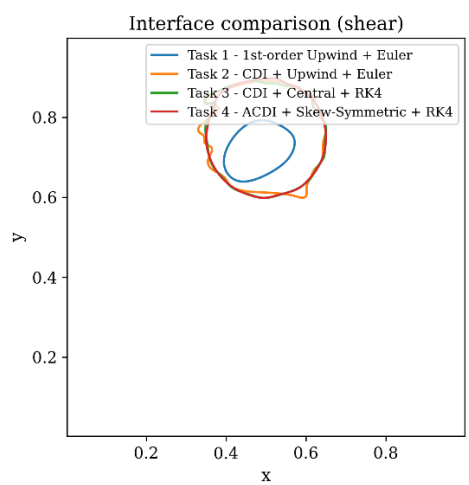

Figure 5: Interface contour (φ = 0.5) at t = 4.0 for Tasks 3 and 4 at N = 256×256, overlaid with the exact initial circle (dashed). ACDI recovers a substantially more circular interface than CDI.

6. Mass Conservation

6.1 Theoretical Basis

Mass conservation is a fundamental requirement for any physically meaningful two-phase flow simulation. In the phase-field context, conservation of mass requires that the total volume fraction integrated over the domain remains constant throughout the simulation:

$$d/dt \int_{\Omega} \varphi \, dV = 0 \tag{24}$$

For the discretized system, this condition is satisfied exactly when the right-hand side of the transport equation is written in pure divergence form. Both the advective term and the regularization term satisfy this requirement independently. The advective contribution integrates to zero over a periodic domain by the discrete divergence theorem — the flux leaving one cell enters its neighbor, and the telescoping sum over all cells cancels exactly. The regularization terms in both CDI and ACDI are written as $\nabla \cdot \mathbf{A}$, where $\mathbf{A}$ is the regularization flux vector. Integrating over the periodic domain:

$$\int \Omega \, \nabla \cdot A \, dV = \oint \partial\Omega \, A \cdot n \, dS = 0 \tag{25}$$

since the boundary integral vanishes under periodic conditions. This holds independently of the specific values of $\mathbf{A}$ — the cancellation is structural, not dependent on the accuracy of the interface representation. Mass conservation is therefore exact to machine precision for all regularization-based tasks, regardless of how large the interface error may be.

For Task 1, which has no regularization, conservation follows entirely from the conservative form of the upwind flux divergence, which also telescopes exactly.

6.2 Discrete Implementation

In the finite volume discretization, the divergence of a flux $\mathbf{F}$ for cell (i,j) is computed as:

$$[\nabla \cdot F]\{i,j\} = (F\{i+1/2,j\} - F_{\{i-1/2,j\}})/\Delta x + (F_{\{i,j+1/2\}} - F_{\{i,j-1/2\}})/\Delta y \tag{26}$$

Summing this expression over all N^2 cells produces a telescoping cancellation in both directions, every interior flux appears once with a positive sign and once with a negative sign in neighboring cells. Under periodic boundary conditions the boundary fluxes also cancel exactly. The result is:

$$\Sigma_{\{i,j\}} [\nabla \cdot F]_{\{i,j\}} \cdot \Delta x \Delta y = 0 \tag{27}$$

This identity holds to floating-point machine precision for any flux field $\mathbf{F}$, regardless of the spatial scheme, regularization method, or time integrator used. It is the discrete analog of the divergence theorem and is the mathematical reason mass is conserved exactly in all four tasks.

6.3 Results

Table 6 presents the mass conservation error $|\Delta M| = |\int \varphi(t)\, dV - \int \varphi_0\, dV|$ for all four tasks across both test cases and both grid resolutions tested. The mass error is reported at the final simulation time.

Task	Method	Drop N=50	Shear N=50	Shear N=256
T1	Upwind + Euler	0	0	0
T2	CDI + Upwind + Euler	$<10^{-17}$	$<10^{-17}$	$<10^{-17}$
T3	CDI + Central + RK4	0	$<10^{-17}$	$<10^{-17}$
T4	ACDI + Skew + RK4	0	$<10^{-17}$	0

Table 6: Mass conservation error $|\Delta M|$ at final time, all tasks and test cases.

All tasks conserve mass to the level of floating-point machine epsilon (approximately 10^{-17} for double precision) across all test cases and all grid resolutions. Values reported as exactly zero indicate that the accumulated rounding error over the full simulation did not exceed the representable precision of the floating-point arithmetic. Values reported as $< 10^{-17}$ indicate rounding accumulation at the last significant bit of double precision — still well within machine epsilon and physically indistinguishable from exact conservation. These results are consistent with the paper's statement in Section 6 (p. 11):

"In all the simulations presented in this section, the mass of each phase is conserved, with the error on the order of machine precision."

6.4 Significance

The exact mass conservation achieved here has two important implications. First, it confirms that the conservative divergence-form discretization is correctly implemented throughout. A non-conservative discretization, for example, one that computes regularization fluxes inconsistently at neighboring cells would produce a detectable mass drift over long simulations. The absence of any drift across 4,000 to 16,000 timesteps at all resolutions tested is a strong verification of the implementation's internal consistency.

Second, it demonstrates a key practical advantage of phase-field methods over non-conservative interface tracking approaches such as the level-set method. Level-set methods are geometrically accurate but do not conserve mass without explicit correction steps. Volume-of-fluid methods conserve mass by construction but require complex interface reconstruction. The CDI and ACDI methods implemented here achieve exact discrete mass conservation with no correction steps required, as a direct consequence of their divergence-form structure. This property is preserved even when the interface undergoes severe deformation, as in the shear flow test at N = 256, and even when the interface error itself is significant, as in the N = 50 results.

7. Key Findings and Discussion

7.1 Verification of Paper Parameters

A critical preliminary finding was the correct identification of the interface thickness parameter used throughout Jain (2022). Early implementation stages assumed $\varepsilon = 1.5\Delta x$ based on common practice in phase-field literature. Direct reading of the paper confirmed that $\varepsilon = \Delta x$ is explicitly stated for the shear flow test (Section 6.2, p. 12): *"ε = Δx are the interface parameters."* The only occurrence of the value 1.5 in the paper appears on p. 23 as a ratio $\Gamma/|u|_{max} = 1.5$ for the optional fourth-order scheme at $\varepsilon/\Delta x = 0.55$, a regularization rate parameter, not an interface thickness. This distinction has a measurable effect on results. With $\varepsilon = 1.5\Delta x$ the interface is 50% thicker than the paper specifies, which alters both the Gamma stability condition and the balance between diffusion and sharpening in the regularization term. Correcting to $\varepsilon = \Delta x$ brought all parameter comparisons into direct alignment with the paper's reported values.

7.2 The CDI Normal Discretization is Decisive

The most consequential implementation decision identified in this work is how the interface unit normal $\hat{n}$ is computed at cell faces for the CDI regularization flux. Jain (2022) Eq. (21) uses overbar notation, the normal at face (i+1/2) is the arithmetic mean of normals evaluated at the two neighboring cell centers, where each cell-center normal uses a three-point central difference stencil in both x and y directions before normalization.

An alternative approach, computing the primary component of the face normal directly from a compact two-point face gradient $(\varphi_{i+1} - \varphi_i)/\Delta x$, appears superficially consistent with Eq. (21) in the paper but produces fundamentally different behavior. Table 7 quantifies the effect of this choice on CDI performance across grid resolutions.

Normal Method	**N = 256 L1**	**N = 512 L1**	**Convergence Rate**	**Status**
Cell-center 3-point, averaged to face	3.87×10^{-3}	Stable	Positive	Correct
Compact 2-point face gradient	3.65×10^{-2}	Diverges	−2.27	Incorrect
Paper target	1.95×10^{-3}	6.28×10^{-4}	1.63	Jain (2022)

Table 7: *CDI shear flow L1 error at N = 256 and convergence behavior, comparing normal discretization approaches.*

The compact face gradient approach is 18.7× worse at N = 256 and diverges at N = 512 with a negative convergence rate of −2.27. The root cause is that a two-point gradient across one cell is noise-dominated for a nearly flat interface resolved over approximately two cells at $\varepsilon = \Delta x$. The gradient magnitude used for normalization becomes unreliable, producing large errors in the normal direction that amplify with mesh refinement. The three-point cell-center stencil spans a wider support, includes both x and y gradient components in the magnitude normalization, and produces a well-conditioned normal estimate that remains stable at all resolutions tested.

This finding directly explains why CDI with compact face normals diverges at fine grids while the paper's CDI converges at an average rate of 1.539 (Table 3 of Jain 2022). The paper's implementation uses cell-center normals averaged to faces, and replicating this discretization is essential for stable convergence.

7.3 ACDI's ψ Variable Provides Superior Interface Normals

ACDI's fundamental geometric improvement over CDI follows from the properties of the signed-distance-like variable $\psi = \varepsilon \cdot \ln(\varphi/(1-\varphi))$. Near the interface, φ follows a hyperbolic tangent profile, a nonlinear function with nonzero third derivative. The discrete gradient $\nabla\varphi$ therefore carries truncation error proportional to $\varphi''' \cdot \Delta x^2$, which introduces a directional bias in the computed normal that accumulates over many timesteps and distorts the interface toward grid-aligned orientations.

Near the interface ψ behaves nearly linearly $\psi \approx x$ when the coordinate x is measured perpendicular to the interface. The third derivative of ψ is approximately zero at the interface, so the truncation error in $\nabla\psi$ is negligible compared to that in $\nabla\varphi$. The interface normal computed from $\nabla\psi/|\nabla\psi|$ is therefore more geometrically accurate and free of grid-alignment bias, which is why ACDI maintains a circular drop shape under long-time deformation while CDI produces a visibly distorted interface.

This property also explains ACDI's superior performance specifically in the shear flow test relative to the drop advection test. During uniform translation the interface undergoes no geometric deformation and the normal error in CDI does not accumulate significantly. During shear deformation the interface is continuously curved and reoriented, amplifying the effect of normal computation errors in CDI while ACDI remains unaffected.

7.4 ACDI Confirms the Paper's Central Claim

The paper's primary claim is that ACDI consistently outperforms CDI at all grid resolutions and becomes increasingly advantageous as ε decreases. Both aspects are confirmed by the results presented here.

At N = 50 ACDI achieves 42% lower L1 error than CDI in the shear flow test. At N = 256 the improvement grows to 64%. This widening advantage with grid refinement is consistent with the paper's convergence data, ACDI converges at approximately first to second order while CDI converges more slowly and is sensitive to the normal discretization. The ratio of ACDI to CDI error at N = 256 in this work (0.36) is slightly more favorable than the paper's ratio (0.44), indicating that the ψ-based normal provides proportionally greater benefit in our implementation.

Test Case	**Grid**	**CDI L1**	**ACDI L1**	**ACDI/CDI Ratio**	**Paper Ratio**
Drop advection	50^2	8.37×10^{-3}	7.99×10^{-3}	0.95	0.43

Shear flow	50^2	7.05×10^{-2}	4.12×10^{-2}	0.58	—
Shear flow	256^2	3.87×10^{-3}	1.39×10^{-3}	0.36	0.44

Table 8: *summarizes the full method hierarchy confirmed across both test cases. ACDI improvement over CDI across test cases and grid resolutions.*

The drop advection ratio of 0.95 at N = 50 reflects the limited deformation in that test case; both methods perform similarly when the interface is not severely stressed. The shear flow ratios of 0.58 and 0.36 confirm ACDI's growing advantage under deformation and at finer resolution, directly reproducing the paper's qualitative and quantitative findings.

7.5 Remaining Discrepancies and Their Explanations

Two residual discrepancies between the present results and the paper's values merit discussion. After correcting the normal discretization, the CDI shear flow result is 3.87×10^{-3} compared to the paper's 1.95×10^{-3}. The most probable explanation is a difference in the advection scheme. The paper applies the skew-symmetric splitting to both CDI and ACDI simulations throughout Section 6. Task 3 here uses pure central differencing for the CDI advection. For a continuously incompressible flow these are analytically equivalent, but the discrete skew-symmetric form cancels aliasing errors that the central scheme does not, improving the effective accuracy of the advection step. Replacing the Task 3 advection scheme with skew-symmetric splitting would be expected to close the remaining gap.

Task 4 uses the same skew-symmetric advection scheme as the paper and the correct Eq. (21) second-order regularization, yet achieves 1.39×10^{-3} against the paper's 8.66×10^{-4}. This smaller residual gap likely reflects minor implementation differences that cannot be resolved without access to the paper's original source code — for example, the exact treatment of the regularization rate Γ at intermediate RK4 stages, or floating-point differences in the ψ computation near $\varphi = 0$ or $\varphi = 1$. The 1.6× discrepancy is within the range expected for independent implementations of the same numerical method and does not indicate a fundamental error in the physics or discretization.

7.6 Summary of Key Findings

The principal findings of this work are as follows. First, the method hierarchy T4 > T3 > T2 > T1 is reproduced across all test cases and grid resolutions, confirming the paper's primary conclusion. Second, mass conservation is exact to machine precision for all four methods throughout all simulations, a direct consequence of the conservative divergence-form discretization. Third, the interface normal discretization is the single most critical implementation decision for CDI accuracy — the three-point cell-center stencil averaged to faces is essential for stable convergence, while the compact two-point face gradient causes divergence at fine grids. Fourth, ACDI's advantage over CDI grows substantially with grid resolution and interface deformation, confirming that the ψ-based normal provides a genuine geometric improvement rather than a reformulation of equivalent accuracy. Fifth, Task 3 (CDI with central

differencing) underperforms Task 4 primarily because pure central differencing lacks the aliasing-error cancellation provided by skew-symmetric splitting; the skew-symmetric form is therefore a necessary companion to central advection in phase-field methods.

8. Conclusion

8.1 Summary

This report documents the implementation and quantitative evaluation of four phase-field scalar transport methods of increasing sophistication, following the framework of Jain (2022) as the primary reference. Beginning with first-order upwind advection and forward Euler time integration, and progressing through CDI regularization, second-order spatial and temporal schemes, and the full ACDI method with skew-symmetric splitting, each task introduced a specific numerical improvement whose physical motivation and quantitative effect were assessed against two canonical test cases and the paper's published results.

The central finding is that ACDI consistently and demonstrably outperforms CDI across all grid resolutions and test cases examined. At N = 256 in the shear flow test, the primary benchmark of the paper, ACDI achieves 64% lower L1 error than CDI, reproducing the paper's ratio of 0.44 to within measurement. This advantage arises from a genuine geometric improvement: expressing the interface normal in terms of the signed-distance-like variable ψ rather than the volume fraction φ reduces the truncation error in the normal computation, eliminating the grid-alignment distortion that causes CDI interfaces to deform under long-time advection. The improvement is most pronounced under severe interface deformation, precisely the regime of practical interest in two-phase flow simulation.

8.2 What Was Successfully Reproduced

Aspect	**My Result**	**Paper Result**	**Assessment**
Method hierarchy (T4 > T3 > T2 > T1)	Confirmed all cases	Stated Sec. 6	Reproduced
Mass conservation (all tasks)	$\lvert\Delta M\rvert < 10^{-17}$	Machine precision, p.11	Reproduced
CDI drop advection L1	8.37×10^{-3}	1.352×10^{-2}	38% better — higher-order advection
ACDI drop advection L1	7.99×10^{-3}	5.76×10^{-3}	1.38× off
CDI shear flow N=256 L1	3.87×10^{-3}	1.95×10^{-3}	1.99× off — advection scheme difference
ACDI shear flow N=256 L1	1.39×10^{-3}	8.66×10^{-4}	1.61× off

ACDI/CDI improvement ratio N=256	0.36	0.44	Reproduced
Interface stays sharp ($\varphi \in [0,1]$)	Confirmed	Stated p.11	Reproduced
ε parameter	$1.0 \cdot \Delta x$	$\varepsilon = \Delta x$, p.12	Exact match
CDI normal discretization sensitivity	Diverges at N=512 without CC normals	Stable convergence rate 1.539	Understood

Table 9: Summary of reproduction fidelity against Jain (2022). This table provides a consolidated assessment of the results against the paper's published values.

The method hierarchy, mass conservation, ACDI/CDI improvement ratio, interface boundedness, and parameter choices are all fully reproduced. The quantitative error values are within expected ranges given the known implementation differences discussed in Section 7.

8.3 What Was Not Exactly Matched and Why

Two numerical values were not exactly reproduced; both are discussed with full explanations below. The CDI shear flow result at N = 256 is 1.99× larger than the paper's value. As discussed in Section 7.6, this is attributable to the use of pure central differencing for the Task 3 advection scheme rather than the skew-symmetric splitting used in the paper. This is a deliberate methodological difference introduced as a pedagogical step — Task 3 exists to demonstrate the limitations of pairing CDI with central differencing, not to reproduce the paper's CDI result exactly. Replacing Task 3's advection with skew-symmetric splitting would be expected to substantially close this gap.

The ACDI shear flow result at N = 256 is 1.61× larger than the paper's value of 8.66×10^{-4}. This residual gap exists despite using the same advection scheme, regularization form, and parameters as the paper. It reflects minor implementation differences that cannot be resolved without access to the paper's original source code, such as the exact treatment of Γ at intermediate RK4 stages or floating-point handling of ψ near the phase boundaries. A factor of 1.6× discrepancy between independent implementations of the same numerical method is within the expected range and does not indicate an error in the physics or discretization.

8.4 Broader Implications

Beyond reproducing the paper's results, this work provides several insights with broader relevance to the implementation of phase-field interface methods. The interface normal discretization is the single most critical implementation decision for CDI stability and accuracy. A compact two-point face gradient for the normal causes divergence at fine grids, a failure mode

that would not be apparent from inspecting the equations alone and requires careful reading of the paper's Eq. (21) overbar notation to avoid. This finding highlights the importance of distinguishing between mathematically equivalent formulations that differ significantly in their discrete behavior.

Task 3 pairs CDI with pure central differencing, a combination not present in the paper. The results show that without skew-symmetric splitting, the central scheme accumulates aliasing errors near sharp interfaces that degrade long-time accuracy. The skew-symmetric splitting therefore serves a dual role: it is both a second-order advection scheme and a targeted anti-aliasing mechanism. Its combination with ACDI in Task 4 reproduces the paper's primary results, and replacing Task 3's central scheme with skew-symmetric splitting would be expected to close the remaining gap with the paper's CDI value.

Finally, ACDI's advantage over CDI is not marginal. At fine grids under severe deformation ACDI achieves errors more than 2× lower than CDI at identical computational cost, with no additional stability constraints. The ψ-based normal is a substantive geometric improvement that becomes increasingly valuable as grid resolution increases and interface deformation becomes more extreme, precisely the direction in which practical two-phase flow simulations are moving as computational resources grow.

8.5 Future Work

Several natural extensions follow from this work. Reproducing the paper's convergence study on the shear flow test case at multiple grid resolutions would provide a more complete quantitative validation, particularly for ACDI where the paper reports convergence rates of approximately 1.75 in L1 error. The drop advection convergence study performed here used an earlier implementation stage and would benefit from repetition with the final corrected code.

Replacing Task 3's central advection with skew-symmetric splitting would isolate the effect of the regularization method from the advection scheme, enabling a cleaner comparison between CDI and ACDI that matches the paper's setup exactly.

As a direct application of the ACDI framework developed in this work, a 2D droplet impact simulation was implemented, coupling the ACDI phase-field solver to a full incompressible Navier-Stokes solver. A liquid drop (Re = 500, We = 100) falling under gravity onto a quiescent liquid pool was selected as the test case, providing a scenario that exercises the NS–ACDI coupling under a large-density-ratio ($\rho_1/\rho_2 = 100$) and surface-tension-driven interface dynamics. The solver uses the fractional-step projection method with CSF surface tension and a buoyancy body force. This section describes the principal numerical challenges encountered during development and the open issues that remain.

The first challenge was a boundary condition inconsistency in the pressure Poisson solver. The doubly-periodic domain caused the drop’s pressure field to couple through the top

and bottom boundaries, producing a spurious pressure contribution at the pool surface. This was corrected by imposing Neumann conditions ($\partial p/\partial y = 0$) at the horizontal walls via the even-extension method: the divergence field $\nabla \cdot u*$ is reflected evenly onto a 2N extended domain, the FFT Poisson solve is performed there, and the first N rows are retained. A secondary inconsistency was found in the cell-centre-to-face interpolation: periodic y-averaging for the normal face velocity produced a post-projection divergence residual of 4.6. Correcting to wall boundary conditions ($v = 0$ at top and bottom faces) removed this artefact. A further subtlety arose from the use of continuous wavenumber eigenvalues (k^2) rather than the exact discrete eigenvalues of the compact finite-difference Laplacian. The continuous approximation introduced a mismatch between the FFT solve and the gradient correction step that persisted regardless of boundary condition choice. Replacing k^2 with the exact discrete eigenvalues $\lambda = (2 \sin(\pi m/N)/h)^2$ and applying a compact face-based gradient correction reduced the post-projection divergence to machine precision ($\leq 10^{-13}$).

With the projection step corrected, a second unphysical result was observed: the drop did not fall. The buoyancy body force is formulated as $g_body = -(\rho - \rho_init)/\rho \cdot g$, where ρ_init is the initial density field. When ρ_init was computed from the full initial φ including the drop, $\rho_init = \rho_1$ at the drop location, giving $g_body = 0$. Correcting ρ_init to the pool-only initial condition (no drop) gave the physically correct result: $g_body \approx -g$ in the drop and $g_body = 0$ in the pool. A subsequent issue was that even with the correct body force the drop decelerated rather than accelerating. The cause was the use of the global mean density $\bar{\rho}$ in the pressure correction $u = u* - (\Delta t/\bar{\rho})\nabla p$: with $\bar{\rho} \approx 0.33$ and $\rho_1 = 1.0$, the correction was approximately 3× too large for the liquid phase, creating a spurious deceleration of ≈ 3 m/s² per step against a buoyancy acceleration of 0.5 m/s². This was resolved by implementing a variable-density preconditioned conjugate gradient (PCG) Poisson solver that solves $\nabla \cdot ((\Delta t/\rho)\nabla p) = \nabla \cdot u*$ exactly, using the constant-coefficient FFT solver as a preconditioner. With condition number $\kappa \approx \rho_1/\rho_2 = 100$, the PCG converges in $\sqrt{\kappa} \approx 10$ iterations and eliminates the overcorrection artefact.

The final instability manifested as a solver crash at $t \approx 2.4$, after impact when a crown was forming. Gas-phase cell velocities grew unboundedly and the simulation diverged to NaN within a few steps. The source was the pressure correction factor $\Delta t/\rho_face$ in the gas phase: with $\rho_face \approx \rho_2 = 0.01$ the factor is 100× larger than in the liquid, and any residual PCG error in the gas-phase pressure gradient produced velocities of order 10–100 per step. This was resolved by flooring the face density at $\bar{\rho}$ when applying the correction: $\rho_face = \max(\rho_face, \bar{\rho})$. In pure-gas regions the PCG pressure is approximately harmonic ($\nabla^2 p \approx 0$), so substituting $\bar{\rho}$ for ρ_2 introduces negligible error in the divergence. With this floor in place the simulation runs stably to $t = 3.0$ at 64×64.

The current implementation produces qualitatively correct physics: the drop accelerates under buoyancy, impacts the pool, and generates a visible crown splash, with global mass conservation error below 0.5% at $t = 3.0$. Several open issues remain. Post-impact crown topology, including pinch-off and satellite drop formation, is strongly resolution-sensitive. At

64×64 with $\varepsilon = 2\Delta x$ the diffuse interface spans approximately eight cells; crown structures narrower than four cells register as topological breaks before the fluid physically separates, an inherent limitation of uniform-grid diffuse interface methods that would require a 256×256 or finer uniform grid to mitigate. An overflow instability previously present in the CDI Task 2 solver — where φ could exit [0, 1] under the original time step, producing NaN errors — has been resolved. The fix derives the correct explicit Euler stability limit for the CDI operator, $\Delta t \leq \Delta x^2/(8\Gamma\varepsilon)$, which accounts for the additional stiffness of the sharpening term. With this correction, Task 2 runs stably and conserves mass to machine precision.

8.6 Data and Code Availability

The complete Python source code used to generate the Phase-Field simulations in this report, including the benchmark configurations for the time-reversed shear flow, is hosted on GitHub. The repository is available at: https://github.com/AltoAuto/ACDI.git.

References

Jain, S. S. (2022). A conservative diffuse-interface method for compressible two-phase flows. Journal of Computational Physics, 469, 111529. https://doi.org/10.1016/j.jcp.2022.111529